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# The onset of turbulence in collisionless magnetic reconnection

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**Abstract.** Two dimensional simulations of collisionless magnetic reconnection produce strong gradients in the current and plasma profiles near the magnetic X-line and along the separatrix during the nonlinear phase. Here we present 3D two-fluid MHD simulations of reconnection that show these sharp gradients break up due to the onset of secondary instabilities, leading to a strongly turbulent configuration in the full 3D system. Two main secondary modes are observed, an electron shear flow instability and the lower-hybrid drift instability. The fluctuations driven by the former mode produce an effective anomalous viscosity (but no resistivity) near the X-line that breaks the frozen-in condition and broadens the main current profile. The lower-hybrid drift fluctuations cause a substantial relaxation of the gradients near the separatrix through a new mechanism, based not on direct particle transport (which is negligible) but rather on the turbulent diffusion of the strong electron flows near the separatrix.

## Introduction

Recent studies (see, eg, [*Birn, et al.*, 2000]) have led to a new understanding of collisionless magnetic reconnection in which the Hall term ($\propto \vec{J}\times\vec{B}$) in the generalized Ohm's law plays a fundamental role. The bulk of this work, however, has been carried out in the context of 2D models, which do not describe instabilities that propagate transverse to the plane of reconnection, *e. g.* those that propagate along the equilibrium current. Two examples of such modes thought to be important in real 3D reconnecting systems are the lower-hybrid drift instability [*Huba, et al.*, 1977] and the electron shear flow instability [*Drake, et al.*, 1994, 1997]. These instabilities are expected to arise naturally from the steep gradients that develop during the nonlinear reconnection process - density or pressure gradients on the $d_i \equiv c/\omega_{pi}$ scale in the case of the lower-hybrid drift instability and current gradients on the $d_e \equiv c/\omega_{pe}$ scale for the electron shear flow mode. Both modes require electron inertia to be unstable, exhibit typical wavelengths comparable to the electron skin depth $d_e$ (for $\beta \sim 1$), and grow on time scales short compared to $1/\omega_{ci}$. We explore here the impact of these small scale instabilities on the collisionless reconnection process with high resolution ($m_i/m_e = 277$) three dimensional simulations of simplest model containing the Hall physics essential to fast reconnection: two-fluid MHD. We find these modes are indeed strongly driven by the profile gradients that are self-consistently generated during reconnection, and rapidly lead to a turbulent outflow region that greatly contrasts the stable, laminar configurations typical of 2D simulations.

The prediction of such turbulent flows is consistent with the substantial body of observational evidence of broadband electric and magnetic fluctuations at the magnetopause and magnetotail (see, *e. g.* [*Gurnett, et al.*, 1979; *Huba, et al.*, 1977]). The lower-hybrid drift instability [*Huba, et al.*, 1977; *Gary and Sgro*, 1990; *Horiuchi and Sato*, 1999] and the electron shear flow mode [*Drake, et al.*, 1994, 1997] have long been regarded as a possible drivers of such turbulence. In the case of the lower-hybrid drift (LHD) instability, the good separation of scales achieved in the 3D simulations described here allow us to go beyond previous work and study the spontaneous onset of the LHD instability during the deep nonlinear phase of reconnection, in which the configuration of the system is no longer dominated by the initial conditions and is no longer well approximated by a simple 1D current layer. In the case of the electron shear flow (ESF) mode, the present study extends past work to address systems with macroscopic scales exceeding $c/\omega_{pi}$ in which the ion motion plays an important role in the reconnection process, and addresses the scaling behavior of the turbulent current layer width in the presence of the ion dynamics.

This work breaks new ground on the anomalous transport generated by the ESF and LHD modes. First, we find the breakdown of the frozen-in condition near the X-line is caused predominantly by an effective anomalous viscosity arising from the ESF fluctuations in the reconnection region. Second, we find the LHD fluctuations cause a substantial relaxation of the gradients near the separatrix through a new mechanism, which is based not on direct particle transport (which is observed to be negligible) but rather on the turbulent diffusion of the strong electron flows near the separatrix. Neither mode is found to produce significant anomalous resistivity or increase the rate of reconnection (rather, a slight slow-down is observed).

It must be emphasized that the two-fluid model, while providing good numerical efficiency and (at least in the 2D case) a reasonably accurate representation of the gross reconnection dynamics, cannot address some potentially important kinetic effects, such the finite $\beta$ stabilization of the LHD mode due to gradient-$B$ drift resonances [*Davidson, et al.*, 1977], finite Larmor radius effects, or ion two-stream instabilities [*Lee*, 1982]. In the LHD mode case, we attempt to assess the accuracy of the fluid description by comparing the two-fluid results in the 2D limit to full particle simulations.

## Simulations

The two-fluid simulation model used here is the same as that described in our previous 2D work [*Shay et al.*, 2000]. We consider the collisionless (zero resistivity) limit in which reconnection is enabled by finite electron inertia. We address

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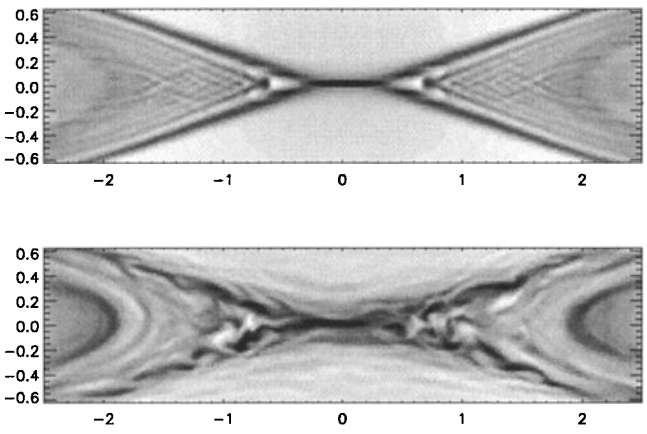



**Figure 1.** Current density in 2D (a) and 3D (b) simulations

the evolution of the periodic system described in [*Shay et al.*, 1999b], which exhibits nearly quasi-steady reconnection. The initial equilibrium is a double current sheet in the $(x, y)$ plane, ie $B_x/B_0 = \tanh[2y/w_0] - \tanh[2(y + L_y/2)/w_0]$ with (unless otherwise noted) a sheet width $w_0 = d_i$, a ion-electron mass ratio $m_i/m_e = 277.8$ (ie $d_e/d_i = 0.06$), a constant total pressure $p + B^2/2 = 1.5B_0^2$, a computational grid $(n_x, n_y, n_z) = (512, 256, 64)$, and a box size $(L_x, L_y, L_z) = (10, 5, 1.25)d_i$. Such a small $L_z$ value is sufficient in the present case because the modes we consider have short ($\sim c/\omega_{pe}$) electron-scale wavelengths in the $z$-direction. Periodic boundary conditions are imposed in all three directions. The initial density and pressure profiles are taken in accordance with the adiabatic law as $p_\alpha/n^\gamma = \text{const}$. We find simulations in the isothermal ($\gamma = 1$) and adiabatic ($\gamma = 5/3$) limits differ only in minor quantitative ways, and for simplicity we report below only the results in the isothermal case with $T_i + T_e = B_0^2/(4\pi n_\infty)$ where $n \simeq n_\infty$ far away from the current sheets. Reconnection is initiated by the addition of a small initial perturbation.

To highlight the role of 3D effects in our simulations, we first review of some past work (*e. g.* [*Shay et al.*, 1999b]) that addressed the 2D limit of the model studied here. In that work, it was shown the magnetic separatrix, together with the out-of-plane current layer and ion flow channel, open up in the downstream direction within a broad, macroscopic X-like region. This is illustrated in Fig. (1a), which shows the out-of-plane current distribution in the vicinity of the magnetic X-line in a 2D simulation with $L_x = 10d_i = 2L_y$, $d_e = 0.06d_i$ ($m_i/m_e = 277.8$) and $B_{0z} = 0$ (the spatial scales are in units of $c/\omega_{pi}$). The magnetic separatrix is nearly coincident with the intense current layer shown in the figure as the dark, narrow X-shaped structure. This current layer is carried by the electrons and exhibits a width comparable to the electron skin depth. It is associated with a similar intensification in the out-of-plane magnetic field along the separatrix (not shown). This structure appears to be essentially a standing whistler wave which, in the rest frame of the inflowing plasma, propagates nearly transverse to $\vec{B}$. Away from the X-line in the downstream direction, the bulk of the out-of-plane current and ion outflow is broadly distributed within the macroscopic regions bounded by the separatrix. The opening-up of the ion flow channel allows the reconnection to proceed at an Alfvenic rate that is independent of the system size and electron mass [*Shay et al.*, 1999b]. The normalized magnetic flux $\psi = \psi_{phys}/(B_0 d_i)$ (with $\vec{B} = \hat{z} \times \nabla\psi_{phys}$) at the X-line vs time in the 2D case is shown in Fig. (2a) as the dashed line ($\psi = 0$ represents the state in which all the flux has been reconnected).

The corresponding result obtained in 3D with a small but finite extent in the out-of-plane direction ($L_z = 1.25$) is shown in Fig. (2a) as the solid line. (On time scales comparable to $1/\omega_{ci}$, inhomogeneities along $z$ average to zero, and one can define a quasi-2D flux function as in the 2D model.) The differences relative to the 2D case arise from the onset of two secondary modes in the 3D simulation: the electron shear flow instability at $t \sim 4$ and the lower-hybrid drift instability at $t \sim 6.5$. Both modes are associated with strong, symmetry breaking ($\partial_z \neq 0$) fluctuations that are not present in the 2D limit. The fluctuations arise predominantly in the out-of-plane current in the case of the electron shear flow instability, while the lower-hybrid drift instability generates perturbations in both the current and plasma density. The onset of these modes can be seen from Fig. (2b), which shows spatial averages $\langle|\partial_z J|\rangle$ (solid line, in units of $B_0/d_i^2$) and $\langle|\partial_z n|\rangle$ (dashed line, with units $n_\infty/d_i$). The nonlinear development of these modes eventually leads to turbulence in the 3D profiles like that shown in Fig. (1b), which is a plot of the out-of-plane current ($t = 6.8$, as in Fig. (1a)). Figs. (3a,b) show the corresponding 2D and 3D plasma density profiles ($t = 7.4$). The comparison of Figs. (2a,b) suggests the onset strong LHD activity at $t \gtrsim 7$ (reflected for example by the rise in the density fluctuations in Fig. (2b)) is associated with a modest ($\sim 25\%$) reduction in the reconnection rate relative to the 2D model. We emphasize these modes arise from the nonlinear development of the system during reconnection, and are stable during the linear phase.

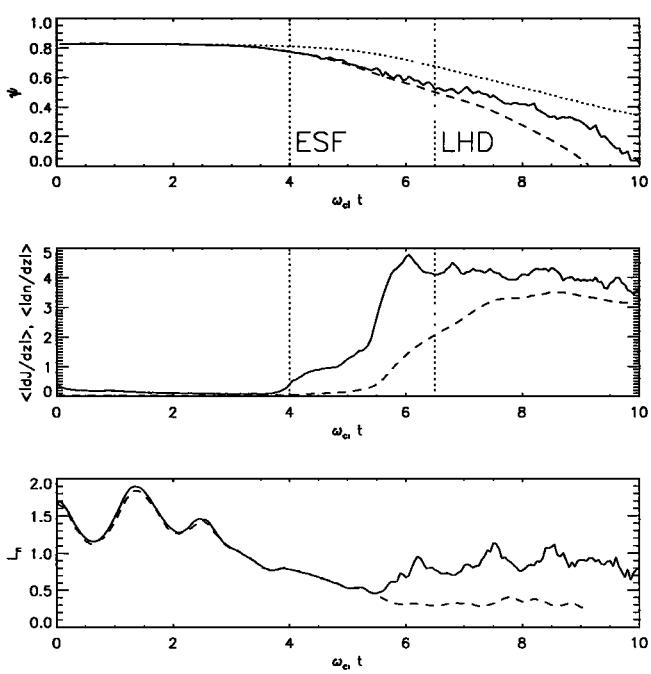



**Figure 2.** (a) Flux vs $t$ in 2D (dashed) and 3d (solid, dotted), (b) current (solid) and density (dashed) fluctuations, (c) density scale length in 2D (dashed) and 3d (solid)

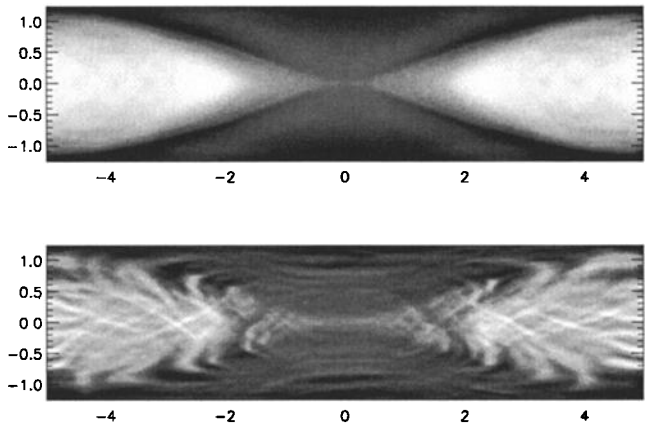



**Figure 3.** Density in 2D (a) and 3D (b) simulations

As an aside, we note similar results have been obtained in a simulation of a sheared configuration with a moderately strong axial field, $B_z = 2B_0$. Although the following discussion will focus on the unsheared case, the reconnected flux vs time obtained in the sheared simulation is plotted in Fig. (2a) as the dotted line.

The electron shear flow instability is mainly driven by the $c/\omega_{pe}$ current layers that form near the magnetic X-line and along the separatrix (see Fig. (1a)). The behavior of this mode is consistent with past studies [*Drake, et al.*, 1994, 1997] based on the electron dynamics alone. It grows on short time scales $\gtrsim 1/\omega_{ce}$, and exhibits $c/\omega_{pe}$-scale wavelengths along the current layer that propagate in the $z$-direction with phase velocities comparable to the out-of-plane electron velocity $V_{ez} \sim J/ne \sim d_e\omega_{ce}$ (assuming $d_eB_x' \sim B_x$). In the presence of the ion dynamics, the propagation of the perturbations along $z$ at such speeds, which greatly exceed the magnetosonic wave velocity in our model, leads to the formation of magnetosonic shock waves that propagate away from current layer into the upstream region. (Such waves produce the density striations visible in Fig. (3b).) We would expect, however, that unless $T_e \gg T_i$, such shock waves would be subject to strong Landau damping in a more realistic, kinetic description.

Comparing the current profile data shown in Figs. (1a,b), one finds the current layers along the separatrix in the 3D case (Fig. (1b)) are about a factor of 2.5 times broader on average that those in the 2D case – that is, about $2.5d_e$ wide compared to only $d_e$. Since $d_i \simeq 16.6d_e$ in this simulation, it is clear the layers in Fig. (1b) remain well below the $c/\omega_{pi}$ scale, suggesting the turbulent current layers along the separatrix in the 3D case continue to scale with $c/\omega_{pe}$ as in the 2D limit. This conclusion is further supported by simulations at $d_e = 0.2d_i$ (a factor of 3.3 increase in $d_e$), which exhibit layers that are about a factor of 3 broader than those in Fig. (1).

Although the ESF mode has little impact on the rate of reconnection, it fundamentally alters the way the frozen-in condition is broken near the magnetic X-line in the simulation. In both the 2D and 3D cases, this condition is relaxed within a $c/\omega_{pe}$-scale region near the X-line by electron inertia, ie the out-of-plane electric field (averaged over $z$ to eliminate the turbulent component in the 3D case) is given by the $z$-component of the electron momentum equation as $\langle E_z\rangle \simeq (m_e/e)\langle \partial_t V_{ez} + \vec{V}_e\cdot\nabla V_{ez}\rangle + \ldots$, where the brackets denote the $z$-average. In the 2D model, the convective term vanishes at the X-line, where $\vec{V}_e = 0$. In the 3D case, however, the ESF fluctuations produce a non-vanishing component of this term that dominates within the reconnection region: $\langle E_z\rangle \simeq (m_e/e)\langle V_{ey}\partial_y V_{ez}\rangle + \ldots$. This is shown in Fig. (4), which in normalized units compares a typical snapshot of $\langle E_z\rangle$ (solid line) to $(m_e/e)\langle V_{ey}\partial_y V_{ez}\rangle$ (dashed) terms. Because any contribution originating from the convective derivative necessarily conserves the total electron momentum, this effect represents an anomalous viscosity and not an anomalous resistivity. Consistent with this interpretation, the integral of the dashed curve in Fig. (4) is nearly zero.

Shortly after the onset of the ESF mode in the simulation of Fig. (1b), the lower-hybrid drift (LHD) instability turns on (at about $t \sim 6.5$). In contrast to the ESF mode, which arises mainly near the X-line where the current gradient is largest, the LHD mode is driven by the density/pressure gradients along the separatrix within the magnetic island like those seen in Fig. (3a). It quickly leads to density fluctuations along the separatrix like those shown in Fig. (3b). These perturbations propagate predominantly across the magnetic field with short wavelengths ($kd_e \simeq 1.2$ initially), and are seen to penetrate roughly half an ion skin-depth into the island. In the saturated state they reach a typical amplitude such that $\tilde{n}'/n' \sim 1$, or equivalently $\tilde{n}/n \sim 1/(kL_n) \sim 0.1$, where $\tilde{n} = n - \langle n\rangle$, $L_n \equiv n/n'$.

The identification of this mode as the LHD instability is based on a comparison of the simulations to the dispersion relation for local modes propagating across the magnetic field in our system. Linearizing our equations about a simple 1D $y$-dependent equilibrium with $\vec{B} = B(y)\hat{x}$, considering modes varying as $\exp(-i\omega t + ikz)$, and applying the equilibrium relation $d(p+B^2/2)/dy = 0$ (or $L_n/L_B = -\beta/2$ with $L_B \equiv B/B'$), one obtains a cubic dispersion relation for local ($kL_n \gg 1$) modes

$$D\omega^2 + kV_A d_i\omega/L_n - k^2V_A^2(1 + D\beta/2) = 0 \qquad (1)$$

where $D(\omega) = 1 + k^2d_e^2(1 + kV_Ad_i/(L_n\omega))$. Some insight can be gained from steep gradient limit $L_n \ll d_i$, in which the highest order term in Eq. (1) can be dropped. Analyzing the resulting quadratic, one obtains unstable modes in the range $(1+\beta/2)L_n/\rho_{si} \lesssim k\rho_{se} \lesssim 1$ where $\rho_{s\alpha}^2 = d_\alpha^2\beta/2$, with a maximum growth rate $\gamma \sim \omega_{lh}$ $(= \sqrt{\omega_{ce}\omega_{ci}})$ and $\omega_r \sim (1+\beta/2)\omega_{ci}L_n/\rho_{se}$. There is no instability for sufficiently weak gradients, $L_n \gg \rho_{si}/(1+\beta/2)$. Applying the dispersion relation to the initial ($t = 0$) current sheet configuration for the simulation of Fig. (3), one finds no instability, consistent with the observed (stable) behavior at early

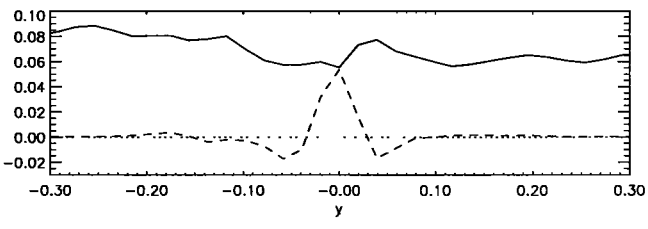



**Figure 4.** Contributions to electric field near X-line

times. If the initial sheet width $w_0$ is reduced from unity to 0.5 (thus reducing $\min(L_n)$ from $1.7d_i$ to $0.85d_i$), however, an instability is predicted, with a maximum growth rate $(\gamma_r, \gamma_i) = (0.55, -2.29)\omega_{lh}$ at $kd_e = 1.20$ for the $y$-value where $L_n$ is minimized. Consistent with this analysis, decreasing $w_0$ to 0.5 in the simulation leads to a rapidly growing mode at early times before the onset of significant reconnection with $kd_e = 1.35$ and $(\gamma_r, \gamma_i) \simeq (0.3, -2.2)\omega_{lh}$. The dependence of the instability on $L_n$ accounts for the onset of the LHD mode at late times in the simulation of Fig. (3). As reconnection progresses in the simulation, the gradients along the separatrix steepen beyond those in the initial state, until finally the LHD instability threshold is crossed during the deep nonlinear phase. This is illustrated in Fig. (2c), which shows the time evolution of the minimum local density scale lengths near the separatrix in the 3D (solid, averaged over $z$) and 2D (dashed) simulations. In the 3D case, the onset of the LHD mode leads to a relaxation of $L_n$ to roughly the linearly marginal value, $L_n \sim \rho_{si} \sim c/\omega_{pi}$.

The simplest mechanism leading to such a relaxation of the density gradient would be direct particle transport driven by the LHD mode. Writing the $z$-averaged continuity equation as $\langle \dot{n} \rangle \simeq -\partial_y \langle n V_y \rangle$, such transport arises from the nonlinear fluctuating contribution $\langle \tilde{n} \tilde{V}_y \rangle$. In agreement with analytic estimates based on linear LHD theory, however, this contribution is found to be negligibly small, and can in fact be completely removed from the simulation with little impact on the relaxation. Rather, the relaxation of the gradient seems to result from the relaxation of the strong in-plane electron currents that arise near the separatrix in the 2D case. Such currents, which are associated with gradients in the out-of-plane field $B_z$ and are the ultimate source of the sharp density gradient near the separatrix [*Shay et al.*, 2000], are strongly diffused by the turbulence (The peak values of $B_z$ in the 3D case, for example, are reduced by 30-40% relative to the 2D limit). This conclusion is supported by numerical tests in which the anomalous transport of $B_z$ was artificially removed, resulting in the continued steepening of the density gradient after LHD onset and the subsequent crash of the simulation. This effect may also account for the modest reduction of the reconnection rate in the 3D case, since it weakens the whistler-wave dynamics in the downstream region that ultimately accelerate the outflowing plasma [*Shay et al.*, 2000].

Finally, we comment on the reproducibility of our LHD results in the context of particle simulations. To study this, we carried out 2D-full particle simulations of a Harris sheet configuration with the same mass ratio used above, $m_i/m_e = 277$. Consistent with the fluid simulation of Fig. (3) at early times, the case $w_0 = 1$ is found to be only very weakly unstable, and the LHD mode has only a slight effect on the initial profile. For $w_0 = 0.5$, however, a strong instability is observed, similar to that in the fluid case, with $(\gamma_r, \gamma_i) \simeq (0.25, -3.8)\omega_{lh}$ at $kd_e = 1.1$. Further comparisons to kinetic simulations in 3D are needed, however, to better assess the physical accuracy of the fluid description used here.

## Conclusion

The two-fluid simulations presented here indicate the sharp gradients that form near the X-line and along the separatrix during collisionless reconnection break up due to the onset of secondary instabilities, leading to a strongly turbulent configuration in the full 3D system. The electron shear flow instability, driven by gradients in the out-of-plane current, leads to a turbulent broadening of the current profile out to scales comparable to a few electron skin depths. This turbulence leads to an anomalous viscosity that balances the reconnection electric field very close to the magnetic X-line. The lower-hybrid drift instability, driven by cross-field gradients in the pressure/density, generates fluctuations in the plasma profiles mainly near the separatrix, and prevents the steepening of the plasma profile scale lengths below about $\rho_{si} = c_s/\omega_{ci}$ for the parameters studied here. It also moderately reduces the time-averaged rate of reconnection.

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